BRAINSTORM GROUP (BSG)

PHY113
General Physics I
E-Test Questions (2022-23)

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1. The comparison of any physical quantity with its standard units is called
   A. Dimension B. Measurement
   C. S.I. Unit D. Unit

2. _____ quantities are all quantities in which laws of physics are described and those measurement is necessary.
   A. Measurement B. Fundamental
   C. Physical D. Derived

3. Physical Quantities that are independent to other quantities are referred to as
   A. Derived Unit
   B. Fundamental Quantities
   C. Basic Unit
   D. Physical Unit

4. The 3 fundamental quantities in mechanics are:
   A. Temperature, Length, Time
   B. Electric Current, Temperature, Intensity
   C. Mass, Time, Length
   D. Mass, Temperature, Time

5. A definite amount of physical quantity is called
   A. Density B. Luminous Intensity
   C. Mass D. Standard Unit

6. Which of the following is a derived Quantity
   A. Density B. Luminous Intensity
   C. Mass D. Electric current

7. Which of the following is a fundamental quantity.
   A. Force B. Power
   C. Mass D. Momentum

8. Which of this not a fundamental S.I Unit
   A. Force B. Kelvin
   C. Mole D. Radian

9. What is the Unit of the following; Length, Breath, Height.
   A. m, s, m B. m, m, m
   C. kg, m, s D. m, kg, s

10. The units of velocity, speed and acceleration are:
    A. m/s, m/s, kg/s
    B. kg/m, m/s, m/s
    C. m/s, m/s, m/s^2
    D. kg, m/s, m/s^2

11. which of the following has unit of meter
    A. Distance B. Lenth
    C. Displacement D. All of the above

12. Dimensional analysis us useful in which of the following way is
    A. To determine if a quantity is fundamental or derived
    B. To determine whether an expression or physical equation is correct or not.
    C. To determine the units of physical quantities

13. How can you describe derived units
    A. Unit
    B. Unit that are dependent on other
    C. None of the above

14. The physical nature of distance is
    A. M B. T
    C. L D. kg

15. Quantities can be added if they have the same
    A. Unit B. Measurement
    C. values D. Dimension

16. The acronyms 'S.I' Unit means
    A. System Intentional
    B. System International
    C. System Intensification
    D. System Identification

17. The base unit of watt is
A. ML^3T^{-3} B. ML^2T^{-3}
C. M^2LT^{-3} D. MLT^{-3}

18. The time rate of increase in velocity is called?
A. Density B. Acceleration
C. Momentum D. Torque

19. The metric prefix for 10^6 is
A. Tera B. Kilo
C. Mega D. Peta

20. The Metric prefix for 10^{-9} is
A. Nano B. Zetta
C. Hecto D. Deka

21. When calculating 'how far' the unit will be in
A. s B. kg
C. m D. s^{-1}

22. When calculating 'how fast' the unit will be in
A. kg B. m/s
C. kg/m D. s

23. Dimension of density, momentum and pressure is
A. MLT, ML^{-3}, MLT^{-2}
B. $ML^{-1}T^{-2}, ML^{-3}, MLT^{-1}$
C. $ML^{-3}, MLT^{-1}, ML^{-1}T^2$
D. $ML^{-3}, ML^{-1}T^{-2}, MLT^{-2}$

24. Dimension of momentum is
A. ML B. ML^2T
C. MLT^{-1} D. MLT

25. What does $10^{-9}F$ means in capacitance
A. Nano farad B. Pico farad
C. Macro farad D. Micro farad

26. The dimension of pressure is
A. ML^2T^{-2} B. $ML^{-1}T^{-2}$
C. MLT^{-1} D. MLT^{-2}

27. The dimension of power is
A. ML^2T^{-3} B. $ML^{-1}T^{-2}$
C. MLT^{-2} D. $ML^{-2}T^{-2}$

28. The dimension of energy is
A. MLT^{-2} B. ML^2T^{-1}
C. ML^2T^{-2} D. MLT^2

29. The product of force and displacement is
A. Pressure B. Work
C. Stress D. Power

30. The acceleration of the falling body is?
A. decreasing B. increasing
C. constant D. varyimg

31. The dimension of volume is
A. M^3 B. kg^3
C. m^2 D. ML^{-3}

32. A vector having magnitude of 5 units north direction is combined with another vector to give zero resultant, what is the vector?
A. 5 units West B. 5 units South
C. 5 units North D. 5 units East

33. If a body move 2m due east and 3m dues west, its displacement is
A. 5 B. -4
C. 0 D. 1

34. Which of the following is not a vector quantity?
A. Direction B. Displacement
C. Momentum D. Force

35. Which of the following is not a vector quantity?
A. force of gravity
B. work, energy & power
C. 2m due east
D. acceleration due to gravity.

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36. Which of the following is a fundamental unit?
A. kg B. kgm^{-3}
C. Nm^{-2} D. ms^{-2}

37. According to Isaac Newton's, space is
A. not empirical B. non-Euclidean
C. absolute D. flat

38. According to Gottfried Leibniz, space was
A. a collection of relations between objects
B. empirical
C. not empirical
D. non-Euclidean

39. The credit of work on visibility of spatial depth is to which philosopher, in the 18th century
A. Gottfried Leibniz
B. George Berkeley
C. Isaac Newton
D. Immanuel Kati.

40. According to Immanuel Kati, space is
A. non-Euclidean
B. flat
C. not empirical
D. absolute

41. In the 19th and 20th centuries mathematician, examine geometric, that are non-Euclidean, in which space is conceived as _____ other than flat.
A. empirical
B. not- empirical
C. curved

42. According to Alert Einstein's theory of general relativity, space around gravitational field, deviate from _____ space.
A. flat
B. non-Euclidean
C. euclidean

43. _____ provide a better model for the shape of spectrum the test of general relatively.
A. Euclidean geometric
B. flat geometric
C. non-Euclidean geometric

44. Time is a scalar quantity
A. True B. False

45. Time is referred to as fundamental quantity
A. True B. False

46. What is the dimension of energy.
A. MLT^{-2} B. ML^2T^{-1}
C. ML^2T^{-2} D. ML^2T^{-3}

47. The rate of change of position is
A. acceleration
B. velocity
C. density
D. momentum

48. Who discovered that space time was curved
A. Immanuel that
B. 19th and 20th centuries mathematician
C. Isaac Newton
D. George Berkeley

49. The independence of space according to sir Isaac Newton is in relation to
(I) those considered of rest
(II) those with constant velocity in a straight line.
A. I and II B. I only
C. II only D. none of the above.

50. The law of physics is the same in very frame of reference is what's theory
A. law of non-relativity
B. Einstein's first postulate
C. Einstein's second postulate
D. None of the above.

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51. Einstein's first postulate is also called
A. principle of initial
B. principle of frame of reference
C. principle relatively
D. none of the above.

52. Goldfried's Leibniz concept of space is contradiction of which of the following concept
A. Immanuel Kati
B. Albert Einstein
C. Isaac Newton
D. Gorge Berkely

53. The physical space has how many dimensions
A. 1 B. 3
C. 5 D. 4

54. The dimension of physical space is
A. length, mass, time
B. length, width, height
C. mass, temperature, height
D. None of the above

55. While newton believed that space absolute, others believed that
A. flat
B. space was defined by the relationship between physical objects
C. space existed independently of any physical objects.

56. When is a frame fixed in the earth an inertial frame
A. what is moving fast
B. when its rotating
C. when the effect of rotation of the earth is negligible
D. when moving slow.

57. When is inertial reference of frame not possible
A. when there is no net force acting on the frame
B. when there is net force acting on the frame
C. A & B
D. none of the above.

58. Special relatively deal, only with physical viewed from
A. principle of relativity
B. Einstein's first postulate
C. Inertial reference frame

59. During the 19th century, most physicist believed that light traveled through a hypothetical medium called _____
A. air B. mercury
C. ether D. optical

60. When is a frame fixed in the earth nit an inertial frame
A. when it is not accelerating
B. when it is moving at a constant speed
C. when it is accelerating

61. Which of the following is not a vector quantity?
A. momentum
B. force
C. temperature
D. work

62. The tendency of a body to be at rest or to move with a constant velocity is known as
A. acceleration
B. law of inertia
C. Newton's 3rd law of motion
D. none of the above.

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63. A scalar quantity is completely specified by
A. number and measurement
B. direction and number
C. displacement and number
D. number and unit

64. The numbers describing the quantities of a particular scalar is known as
A. displacement
B. direction
C. magnitude
D. measurement

65. A vector quantity is completely specified by both
A. magnitude and measurement
B. measurement and direction
C. magnitude and direction
D. all of the above.

66. The type of vector whose line to action can be shifted parallel to is self is
A. localized vector
B. bonded vector
C. free vector
D. unit vector.

67. The type of vector whose line of action can not be shifted parallel to itself is called
A. localized vectors
B. coplanar vector
C. zero vector
D. equal vector

68. The vector which lies in the same plane are called
A. negative vector
B. equal vector
C. coplanar vector
D. unit vector

69. The vector which passes through the common point are called
A. coplanar vector
B. concurrent vector
C. bounded vector

70. The vector which has the same magnitude as the vector 'a' but opposite in direction to 'a' is called
A. equal vector
B. negative vector
C. unit vector
D. bounded vector

71. An example of a free vector is
A. Force
B. Displacement
C. Density
D. velocity

72. Force and momentum are examples of what type of vector.
A. localized vector
B. coplanar
C. concurrent
D. unit vector.

73. The null of zero vector is denoted by
A. [ ] B. ∅
C. 0 D. θ

74. Which of the following has the same dimension of stress?
A. Flow rate B. Pressure
C. Work D. Momentum

75. The derived dimension $[ML^2T^{-2}]$ is a dimension of
I. acceleration
II. Torque
III. energy
A. I only B. III only
C. I and II only D. II and III only

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ANSWERS TO PHY113							
1.	B	21.	C	41.	C	61.	C
2.	C	22.	B	42.	C	62.	B
3.	B	23.	C	43.	C	63.	D
4.	C	24.	C	44.	A	64.	C
5.	D	25.	A	45.	A	65.	A
6.	A	26.	B	46.	C	66.	C
7.	C	27.	A	47.	B	67.	A
8.	D	28.	C	48.	B	68.	C
9.	B	29.	B	49.	A	69.	B
10.	C	30.	C	50.	B	70.	B
11.	D	31.	A	51.	C	71..	B
12.	B	32.	B	52.	C	72.	A
13.	B	33.	D	53.	B	73.	C
14.	C	34.	A	54.	B	74.	B
15.	D	35.	Bs	55.	B	75.	D
16.	B	36.	A	56.	C	76.	
17.	B	37.	C	57.	B	77.	
18.	B	38.	A	58.	C	78.	
19.	C	39.	B	59.	C	79.	
20.	A	40.	C	60.	C	80.	

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